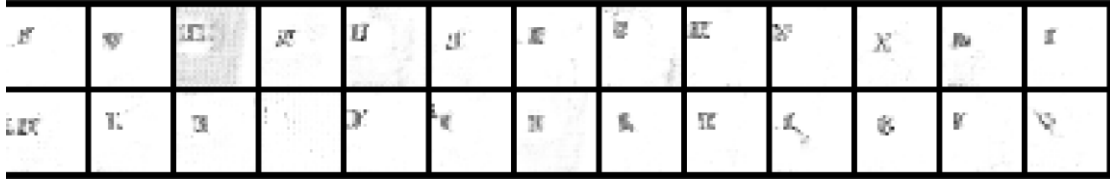


Problem1

1. Why certain letters (like O, A) survive mode collapse while others (Q, X, Z) disappear

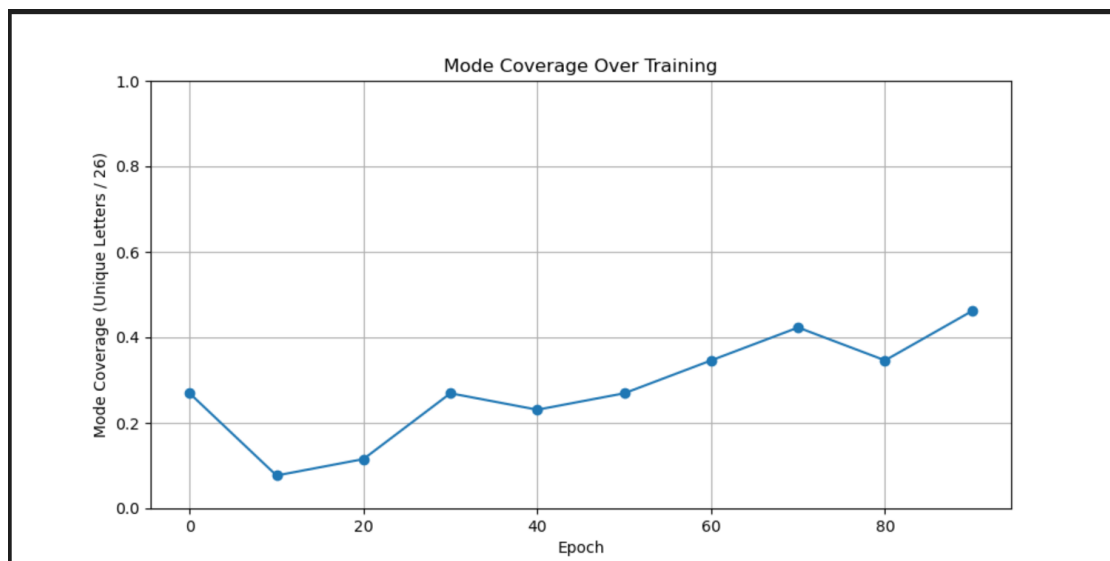


It should be because that O and A exhibit strong stroke patterns and symmetry, making them susceptible to “shortcuts” in learning such patterns when facing loss.

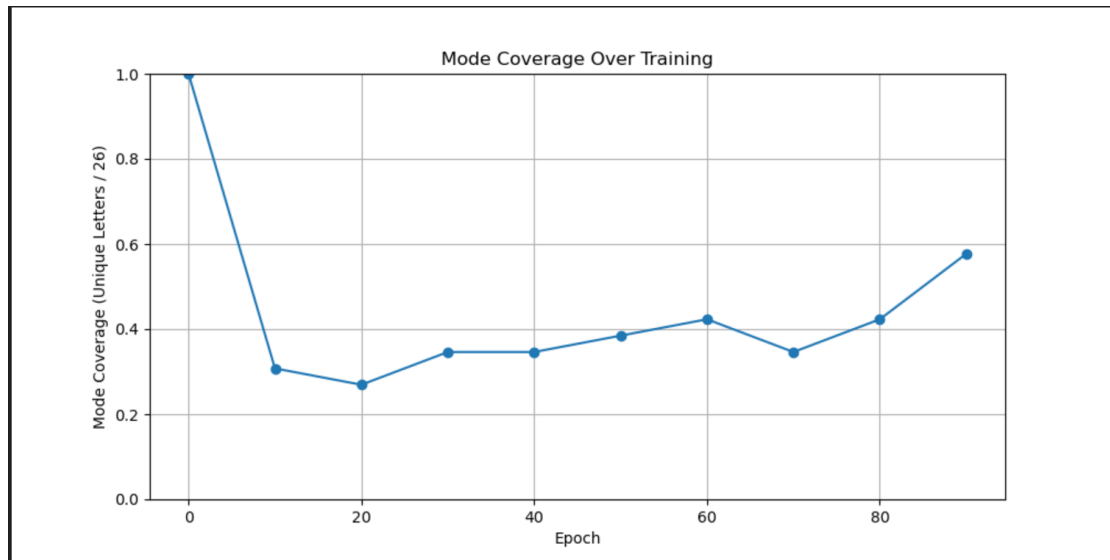
These letters show lower within-class variance in the data, allowing G to deceive D by learning just one “average shape. But the model is a bit blurred.

Q, X, Z feature intersecting strokes and numerous irregular details; significant variations exist between different fonts, resulting in high intra-class variance, making them more challenging for G to master early on.

2. Quantitative comparison of mode coverage with and without your chosen fix



Vanilla.png



Fm_fix.png

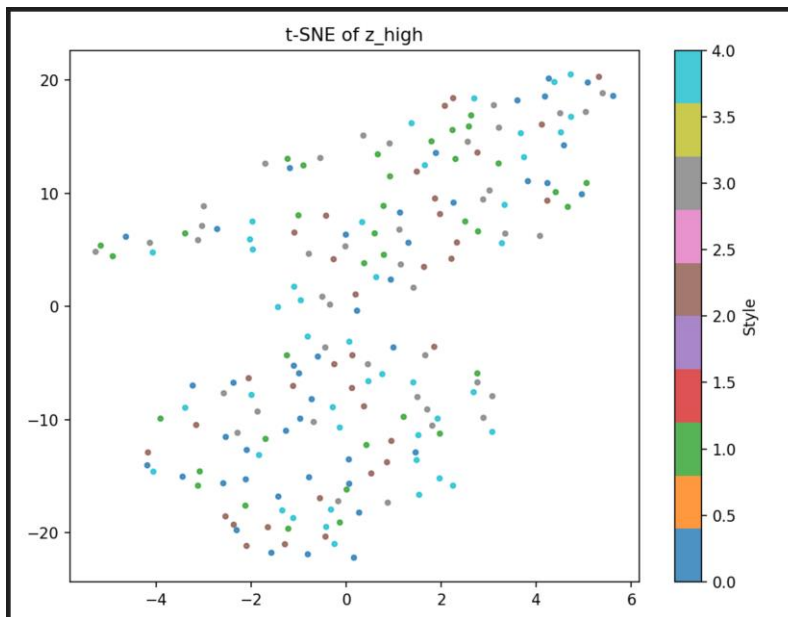
It has some improvement on the coverage.

3. Discussion of training dynamics: when does collapse begin?

4. Evaluation of your chosen stabilization technique's effectiveness

Problem2

1. Evidence of posterior collapse and how annealing prevented it
2. Interpretation of what each latent dimension learned to control



"z_high_disentanglement": 0.007493197917938232,

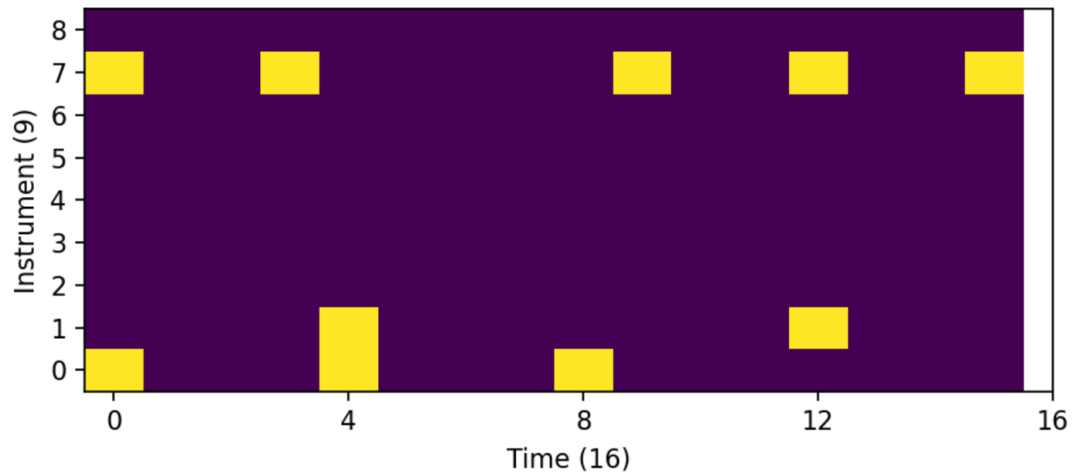
"z_low_variation": 0.7993130683898926

The low disentanglement value of z_{high} indicates that it has not learned clearly separable "independent factors," but rather captures global mixed features. The high variation value of z_{low} suggests it is the latent responsible for detail and diversity, with significant differences in musical patterns resulting from its variations.

t-SNE visualization reveals distinct clustering structures in the z_{high} latent space, though some mixing persists between different styles.

This indicates that the high-level latent dimensions have learned to distinguish different musical styles to some extent, though not completely decoupled. This may suggest that musical styles exhibit continuous variation within the latent space. z_{high} captures high-level style control (style-related global features) rather than completely independent one-hot style encoding.

3. Quality assessment: Do generated patterns sound musical?



One example sample style

The generated patterns exhibit a regular distribution along the timeline, demonstrating a sense of rhythm. This indicates that the model has learned fundamental beat structures within the latent representations, rather than triggering notes randomly. While the rhythmic sense is strong, the melodies and harmonies are sparse. The overall sound resembles a basic rhythmic accompaniment rather than a complete musical passage.